凑微分公式示例及计算结果 (共 10 类)

(1) $\int f(ax+b)dx = \frac{1}{a}\int f(ax+b)d(ax+b)$

1. $\int e^{2x+1}dx = \frac{1}{2}\int e^{2x+1}d(2x+1) = \frac{1}{2}e^{2x+1}+C$
2. $\int \sin(3x-2)dx = \frac{1}{3}\int \sin(3x-2)d(3x-2) = -\frac{1}{3}\cos(3x-2)+C$
3. $\int (4x+5)^6dx = \frac{1}{4}\int (4x+5)^6d(4x+5) = \frac{1}{28}(4x+5)^7+C$
4. $\int \frac{1}{5x+3}dx = \frac{1}{5}\int \frac{1}{5x+3}d(5x+3) = \frac{1}{5}\ln|5x+3|+C$
5. $\int \sec^2(2x)dx = \frac{1}{2}\int \sec^2(2x)d(2x) = \frac{1}{2}\tan(2x)+C$

(2) $\int x^m f(ax^{m+1}+b)dx = \frac{1}{(m+1)a}\int f(ax^{m+1}+b)d(ax^{m+1}+b) \quad (m\neq -1)$

1. $\int xe^{x^2+1}dx = \frac{1}{2}\int e^{x^2+1}d(x^2+1) = \frac{1}{2}e^{x^2+1}+C$
2. $\int x^2\cos(2x^3-1)dx = \frac{1}{6}\int \cos(2x^3-1)d(2x^3-1) = \frac{1}{6}\sin(2x^3-1)+C$
3. $\int x^3\sqrt{3x^4+2}dx = \frac{1}{12}\int \sqrt{3x^4+2}d(3x^4+2) = \frac{1}{18}(3x^4+2)^{\frac{3}{2}}+C$
4. $\int \frac{x}{4x^2+5}dx = \frac{1}{8}\int \frac{1}{4x^2+5}d(4x^2+5) = \frac{1}{8}\ln(4x^2+5)+C$
5. $\int x^4\sin(x^5)dx = \frac{1}{5}\int \sin(x^5)d(x^5) = -\frac{1}{5}\cos(x^5)+C$

(3) $\int f(\sqrt{x})\frac{dx}{\sqrt{x}} = 2\int f(\sqrt{x})d(\sqrt{x})$

1. $\int \frac{\sin(\sqrt{x})}{\sqrt{x}}dx = 2\int \sin(\sqrt{x})d(\sqrt{x}) = -2\cos(\sqrt{x})+C$
2. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}dx = 2\int e^{\sqrt{x}}d(\sqrt{x}) = 2e^{\sqrt{x}}+C$
3. $\int \frac{\cos(\sqrt{x})}{\sqrt{x}}dx = 2\int \cos(\sqrt{x})d(\sqrt{x}) = 2\sin(\sqrt{x})+C$
4. $\int \frac{1}{\sqrt{x}(1+\sqrt{x})}dx = 2\int \frac{1}{1+\sqrt{x}}d(\sqrt{x}) = 2\ln|1+\sqrt{x}|+C$
5. $\int \frac{\ln(\sqrt{x})}{\sqrt{x}}dx = 2\int \ln(\sqrt{x})d(\sqrt{x}) = 2\sqrt{x}(\ln\sqrt{x}-1)+C$

(4) $\int f(e^x)e^xdx = \int f(e^x)d(e^x)$

1. $\int e^x\cos(e^x)dx = \int \cos(e^x)d(e^x) = \sin(e^x)+C$
2. $\int e^x\sin(e^x)dx = \int \sin(e^x)d(e^x) = -\cos(e^x)+C$
3. $\int \frac{e^x}{1+e^{2x}}dx = \int \frac{1}{1+(e^x)^2}d(e^x) = \arctan(e^x)+C$
4. $\int e^x\sqrt{1+e^x}dx = \int \sqrt{1+e^x}d(e^x) = \frac{2}{3}(1+e^x)^{\frac{3}{2}}+C$
5. $\int e^x\ln(e^x)dx = \int \ln(e^x)d(e^x) = e^x(x-1)+C$

(5) $\int f(\ln x)\frac{1}{x}dx = \int f(\ln x)d(\ln x)$

1. $\int \frac{\cos(\ln x)}{x}dx = \int \cos(\ln x)d(\ln x) = \sin(\ln x)+C$
2. $\int \frac{\sin(\ln x)}{x}dx = \int \sin(\ln x)d(\ln x) = -\cos(\ln x)+C$
3. $\int \frac{1}{x(1+\ln^2 x)}dx = \int \frac{1}{1+(\ln x)^2}d(\ln x) = \arctan(\ln x)+C$
4. $\int \frac{\ln^2 x}{x}dx = \int (\ln x)^2d(\ln x) = \frac{1}{3}(\ln x)^3+C$
5. $\int \frac{e^{\ln x}}{x}dx = \int e^{\ln x}d(\ln x) = e^{\ln x}+C = x+C$

(6) $\int f(\sin x)\cos xdx = \int f(\sin x)d(\sin x)$

1. $\int \sin^2 x\cos xdx = \int \sin^2 xd(\sin x) = \frac{1}{3}\sin^3 x+C$

2. $\int e^{\sin x}\cos x dx = \int e^{\sin x} d(\sin x) = e^{\sin x} + C$
3. $\int \cos(\sin x)\cos x dx = \int \cos(\sin x) d(\sin x) = \sin(\sin x) + C$
4. $\int \frac{\cos x}{1+\sin^2 x} dx = \int \frac{1}{1+\sin^2 x} d(\sin x) = \arctan(\sin x) + C$
5. $\int \sqrt{\sin x}\cos x dx = \int \sqrt{\sin x} d(\sin x) = \frac{2}{3}(\sin x)^{\frac{3}{2}} + C$

(7) $\int f(\cos x)\sin x dx = -\int f(\cos x) d(\cos x)$

1. $\int \cos^3 x \sin x dx = -\int \cos^3 x d(\cos x) = -\frac{1}{4}\cos^4 x + C$
2. $\int e^{\cos x}\sin x dx = -\int e^{\cos x} d(\cos x) = -e^{\cos x} + C$
3. $\int \sin(\cos x)\sin x dx = -\int \sin(\cos x) d(\cos x) = \cos(\cos x) + C$
4. $\int \frac{\sin x}{1+\cos^2 x} dx = -\int \frac{1}{1+\cos^2 x} d(\cos x) = -\arctan(\cos x) + C$
5. $\int \sqrt{\cos x}\sin x dx = -\int \sqrt{\cos x} d(\cos x) = -\frac{2}{3}(\cos x)^{\frac{3}{2}} + C$

(8) $\int f(\tan x)\frac{1}{\cos^2 x} dx = \int f(\tan x) d(\tan x)$

1. $\int \tan^3 x \frac{1}{\cos^2 x} dx = \int \tan^3 x d(\tan x) = \frac{1}{4}\tan^4 x + C$
2. $\int e^{\tan x}\frac{1}{\cos^2 x} dx = \int e^{\tan x} d(\tan x) = e^{\tan x} + C$
3. $\int \sin(\tan x)\frac{1}{\cos^2 x} dx = \int \sin(\tan x) d(\tan x) = -\cos(\tan x) + C$
4. $\int \frac{1}{1+\tan^2 x}\frac{1}{\cos^2 x} dx = \int \frac{1}{1+\tan^2 x} d(\tan x) = \arctan(\tan x) + C$
5. $\int \sqrt{\tan x}\frac{1}{\cos^2 x} dx = \int \sqrt{\tan x} d(\tan x) = \frac{2}{3}(\tan x)^{\frac{3}{2}} + C$

(9) $\int f(\arcsin x)\frac{1}{\sqrt{1-x^2}} dx = \int f(\arcsin x) d(\arcsin x)$

1. $\int \frac{(\arcsin x)^2}{\sqrt{1-x^2}} dx = \int (\arcsin x)^2 d(\arcsin x) = \frac{1}{3}(\arcsin x)^3 + C$
2. $\int \frac{e^{\arcsin x}}{\sqrt{1-x^2}} dx = \int e^{\arcsin x} d(\arcsin x) = e^{\arcsin x} + C$
3. $\int \frac{\cos(\arcsin x)}{\sqrt{1-x^2}} dx = \int \cos(\arcsin x) d(\arcsin x) = \sin(\arcsin x) + C = x + C$
4. $\int \frac{1}{\arcsin x\sqrt{1-x^2}} dx = \int \frac{1}{\arcsin x} d(\arcsin x) = \ln|\arcsin x| + C$
5. $\int \frac{\sqrt{\arcsin x}}{\sqrt{1-x^2}} dx = \int \sqrt{\arcsin x} d(\arcsin x) = \frac{2}{3}(\arcsin x)^{\frac{3}{2}} + C$

(10) $\int f(\arctan x)\frac{1}{1+x^2} dx = \int f(\arctan x) d(\arctan x)$

1. $\int \frac{(\arctan x)^3}{1+x^2} dx = \int (\arctan x)^3 d(\arctan x) = \frac{1}{4}(\arctan x)^4 + C$
2. $\int \frac{e^{\arctan x}}{1+x^2} dx = \int e^{\arctan x} d(\arctan x) = e^{\arctan x} + C$
3. $\int \frac{\sin(\arctan x)}{1+x^2} dx = \int \sin(\arctan x) d(\arctan x) = -\cos(\arctan x) + C$
4. $\int \frac{1}{(1+(\arctan x)^2)(1+x^2)} dx = \int \frac{1}{1+(\arctan x)^2} d(\arctan x) = \arctan(\arctan x) + C$
5. $\int \frac{\sqrt{\arctan x}}{1+x^2} dx = \int \sqrt{\arctan x} d(\arctan x) = \frac{2}{3}(\arctan x)^{\frac{3}{2}} + C$